

**Time:** 5 minutes**Outcome:** Read payoff matrices and identify best responses, dominant strategies, and Nash equilibrium.**Difficulty:** ●●○ Intermediate

Game Theory Foundations: Payoff Matrices, Best Responses & Nash Equilibrium



THE CORE IDEA

Game theory represents strategic choices with players, actions, and payoffs that depend on what others do. Correctly reading the matrix is the prerequisite for identifying best responses and equilibrium. Compare mutual best responses with the combined-payoff maximum. A Nash equilibrium is a cell containing mutual best responses.



HOW TO RECOGNIZE IT

- Identify the players, each player's available actions, and the payoff associated with every action combination.
- For each possible rival action, mark the action that gives a player the highest payoff.
- A dominant strategy is best regardless of the rival's action.
- A Nash equilibrium is a cell where every player's action is a best response to the others.



WATCH OUT

Do not assume equilibrium is efficient. Compare mutual best responses with the combined-payoff maximum. A Nash equilibrium is a cell containing mutual best responses.



WORKED EXAMPLE: FINDING A NASH EQUILIBRIUM

Payoff order: (Row firm, Column firm)

	X	Y
A	(9, 9)	(2, 3)
B	(3, 2)	(7, 7)

For each strategy of the other player, mark the action that gives a player the larger payoff. In the displayed matrix, row A against column X gives (9, 9), while switching unilaterally to row B would lower the row player's payoff from 9 to 3. The column player also prefers X to Y when row A is chosen, so (A, X) is a cell of mutual best responses and is a Nash equilibrium.



CHECK YOURSELF

Why must payoff order be stated?

**READY?**

Return to the game and master this concept.